Jinay Patel

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Education

University of Waterloo
Bachelor of Computer Engineering *Expected Graduation: Jun 2031*

Skills

Languages: C/C++ (5 years), JavaScript/TypeScript (6 years), Python (5 years), Java (2 years), Rust (3 years)
Technologies: PyTorch, LangChain, Mastra, CMake, vcpkg, React, Svelte, Next JS, Tailwind CSS, Git, Docker, Unity, Linux

Experience

Lead Software Engineer October 2023 — June 2026
Seaquam Robotics *Delta, BC*
- Built a data backend with Deno and TypeScript to aggregate data from 3 sources, cutting manual data entry by 50%
- Created a custom React and Tauri desktop CMS app, reducing data entry time from weeks to 60 minutes
- Built a Deno backend caching layer for slow third-party API endpoints, increasing server **response speed by 150%**

Embedded Controls Software Engineer October 2021 — June 2023
Seaquam Robotics *Delta, BC*
- Implemented C++ motion algorithms, securing a **100% win rate** during autonomous periods across 15+ matches
- Designed an interpolated lookup table using a monotonic cubic spline to tune PID constants in real time, increasing movement accuracy by 200% with **error tolerances under 1 inch**

Software Engineering Tutor September 2025 — March 2026
Code Ninjas *Surrey, BC*
- Helped 5 senior Black Belt students debug and implement features for their games being built with C# and Unity
- Designed ad campaigns with visual assets for Code Ninjas, improving the local branch's search visibility

Freelance Systems Administrator / IT Consultant
Convergence Media *Remote*
- Created a custom Linux Mint ISO using Cubic with 3 bash scripts to automate the installation and setup of programs
- Deployed a self-hosted Nextcloud server (Microsoft 365 alternative) using Docker, which allowed file sharing and private video calls
- Configured a **Tailscale tunnel** to securely expose the Nextcloud server for remote video calls and file sharing

Projects

Posturai | Hackathon Project Devpost
- A Posture correction app that detects bad posture in real time and warns a user about it, built using Python and Streamlit
- Trained a Machine Learning model using **PyTorch** to classify good and bad posture with a dataset of just 150 images

Lynx (Automated Note Organizer) | Hackathon Project Obsidian Plugin Listing
- Utilized LangChain to automatically generate tags, a title, and references to other notes with LLMs running locally on Ollama
- Launched the plugin publicly on Obsidian's community store during the span of the hackathon and gained **100+ users**
- Built a **RAG + GraphRAG** system for answering user queries and surfacing relevant notes, combining a local Graphology database, LangChain Memory Vector Stores and an embedding model running locally

AI Complaint Categorization System | Personal Project GitHub
- Built a pipeline that analyzes multimodal complaint data and auto-categorizes it using OCR Space, Gemini, and Hugging Face
- Designed a **RAG pipeline** that embeds and stores complaint data in a vector database, then retrieves relevant records via **LangChain**

Awards

Vex Robotics World Champion, RECF & Vex Robotics April 2026
Top overall team among 20,000+ worldwide, recognized for excellence across robot design, programming, and competitive performance

Excellence Award @ Provincial Championship, Vex Robotics February 2026
Awarded to the top team out of the 100+ teams in British Columbia for excellent performance

VSHacks Hackathon 2nd Place, VSHacks July 2025
Awarded for developing Posturai, a real-time posture correction app

Robotics Programming Award x3, Waterloo CEMC May 2026
Awarded for innovative motion control algorithms and technical documentation

Senior Computer Programming Award, Seaquam Secondary School June 2026